

HOUSE PRICE PREDICTION USING LINEAR REGRESSION

Internship Project Report

1. Introduction

Machine Learning is widely used to analyze data and make predictions. In this project, a Linear Regression model was developed to predict house prices using the California Housing Dataset.

The main objective of this project was to understand the complete machine learning workflow, including data loading, exploratory data analysis (EDA), model training, prediction, and evaluation.

This project helped me gain practical experience with Python and machine learning libraries such as Pandas, NumPy, Matplotlib, Seaborn, and Scikit-Learn.

2. Dataset Description

The California Housing Dataset was used for this project. The dataset is available in the Scikit-Learn library and contains information about houses located in different areas of California.

Dataset Information

- Total Records: 20,640
- Total Features: 8
- Target Variable: Price

Features

- MedInc (Median Income)
- HouseAge
- AveRooms
- AveBedrms
- Population
- AveOccup
- Latitude
- Longitude

The target variable is **Price**, which represents the house value.

3. Exploratory Data Analysis (EDA)

Before building the model, the dataset was explored to understand its structure and characteristics.

The following functions were used:

- `df.head()`
- `df.shape`
- `df.info()`
- `df.describe()`

Missing Values Check

The dataset was checked for missing values using the `df.isnull().sum()` function.

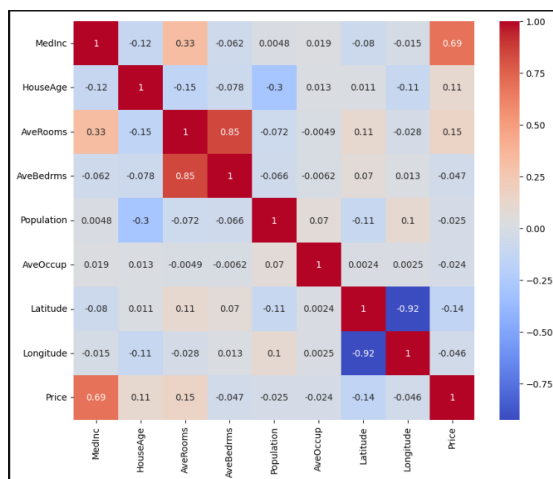
Observation:

No missing values were found in the dataset.

Data Visualization

Different visualizations were created to better understand the data.

Figure 1: Correlation Heatmap



Observation:

The heatmap shows relationships between different features. Median Income (MedInc) has the strongest positive correlation with house prices, making it one of the most important features for prediction.

4. Model Development

For model training, the dataset was divided into input features (X) and target variable (y).

Feature Selection

Input Features:

- MedInc
- HouseAge
- AveRooms
- AveBedrms
- Population
- AveOccup
- Latitude
- Longitude

Target Variable:

- Price

Train-Test Split

The dataset was divided into:

- Training Data: 80%
- Testing Data: 20%

This approach helps evaluate model performance on unseen data.

Algorithm Used

Linear Regression was used as the machine learning algorithm for predicting house prices.

The model was trained using the training dataset and then used to generate predictions on the testing dataset.

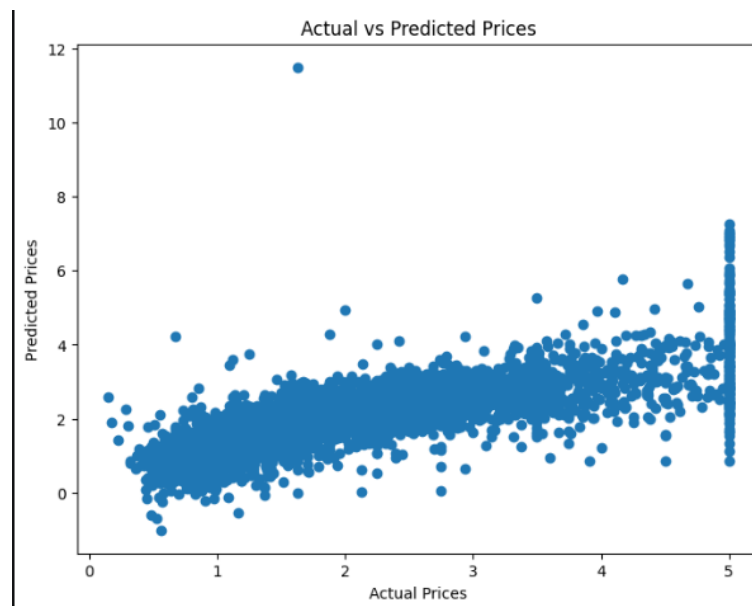
5. Model Evaluation

The model was evaluated using standard regression metrics.

Results

- Mean Absolute Error (MAE): 0.5332
- Mean Squared Error (MSE): 0.5559
- R^2 Score: 0.5758

Figure 2: Actual vs Predicted House Prices



Observation:

The graph shows a positive relationship between actual and predicted values. Most predictions follow the overall trend of actual house prices, indicating reasonable model performance.

Interpretation

The R^2 score of approximately 0.58 indicates that the model explains about 58% of the variation in house prices.

The MAE and MSE values suggest that the model predictions are reasonably close to actual values.

6. Conclusion

In this project, a House Price Prediction model was successfully developed using Linear Regression.

The California Housing Dataset was explored, visualized, and used for training the model. The model achieved satisfactory results and provided a good understanding of the machine learning workflow.

This project helped me learn data analysis, visualization, model training, prediction, and evaluation using Python.

7. Future Improvements

The performance of this project can be improved by:

- Using Random Forest Regression
- Applying XGBoost
- Performing feature engineering
- Hyperparameter tuning
- Testing additional machine learning algorithms

These improvements may increase prediction accuracy and overall model performance.